

Taha Iqbal

Philadelphia, PA

Tahai@seas.upenn.edu • (267) 506-6888 • github.com/Tahai-nformatics • linkedin.com/in/Taha-Iqbal-001

EDUCATION

University of Pennsylvania

Philadelphia, PA

Master of Science in Engineering: Scientific Computing

2026

Certificate in Biomedical Informatics

Cumulative GPA: 3.70/4.00

Temple University

Philadelphia, PA

Professional Science Master's: Bioinformatics

2019

State University of New York at Albany

Albany, NY

Bachelor of Science: Biology

2016

SKILLS

Programming Languages: Python, C++, R, SQL, Java, Bash/Shell

Bioinformatics: NGS (DNA-Seq, RNA-Seq), GWAS, GATK, PLINK, BCF/VCF, Bioconductor, Samtools, Bedtools, Fine-Mapping

Machine Learning: Linear & Logistic Regression, Decision Trees, Random Forest, NLP, scikit-learn, TensorFlow, PyTorch

Cloud & HPC: AWS (S3, EC2, Glacier), SLURM, HPC clusters

Pipelines, Containers, Version Control: Snakemake, Nextflow, Docker, Singularity, Git

Backend & APIs: RESTful API development (Flask, FastAPI)

PROFESSIONAL EXPERIENCE

Perelman School of Medicine, University of Pennsylvania

2021 – Present

Bioinformatician II

- Developed and applied reproducible NGS and GWAS analysis workflows for large-scale genomic datasets, enabling variant discovery, cohort-level association analyses, downstream statistical modeling, and robust validation across 50,000+ whole genomes
- Performed large-scale genomic data analyses including variant QC, cohort harmonization, phenotype integration, rare variant aggregation, burden testing support, and longitudinal disease progression analyses using population-scale sequencing datasets
- Built computational workflows for genomic data processing and analysis, including variant extraction, annotation, filtering, genotype harmonization, and preparation of datasets for downstream machine learning and statistical genetics applications
- Optimized high-throughput bioinformatics workflows on AWS (EC2, S3, Glacier) and HPC clusters using parallelized processing strategies, significantly improving scalability and runtime efficiency for terabyte-scale sequencing analyses
- Managed and harmonized large-scale genomic datasets across multiple cohorts, including genome build liftover (hg18/hg19 to GRCh38), metadata standardization, release validation, and reproducible version-controlled data distribution
- Optimized bioinformatics workflows for execution on AWS and HPC clusters using parallelized and scalable processing strategies for large-scale sequencing analyses
- Collaborated with multidisciplinary teams including computational biologists, statisticians, and investigators to interpret genomic findings, support publication-ready analyses, and develop tailored analytical approaches for ongoing research studies
- Mentored junior researchers and lab members on genomic data analysis, pipeline development, statistical workflows, and reproducible computational research practices

Aperiomics, Sterling, VA

2020 – 2021

Clinical Bioinformatician

- Designed and deployed a NGS pipeline for *Human Papillomavirus* (HPV), enabling accurate identification of Alpha, Beta, and Gamma strains for NIH and X4 Pharmaceuticals through analysis of FASTQ, BAM/CRAM, and aligned sequencing data
- Simulated genomic reads and aligned closely related HPV genomes using APIs and custom workflows, establishing thresholds for strain detection and limits of identification
- Processed and analyzed sequencing data from clinical samples (fecal, plasma, urine, CSF, tissue) by aligning FASTQ and BAM/CRAM files against a curated database of 40,000 bacterial, viral, and fungal genomes for pathogen identification and clinical reporting
- Automated data transfer, sequencing data processing, and report/variant filter generation workflows, significantly improving turnaround time for results delivered to clinicians and patients
- Implemented containerized bioinformatics workflows using Docker and executed scalable pipelines on AWS (S3, EC2, Glacier) and local HPC clusters to ensure reproducibility, scalability, and secure data access
- Conducted validation studies for fecal and plasma sequencing assays to evaluate pipeline performance, sensitivity, and authored SOPs and QC documentation in compliance with CAP regulations to maintain clinical testing standards

PUBLICATIONS

- Rajabli, F., Benchek, P., Tosto, G. et al. Multi-ancestry genome-wide meta-analysis of 56,241 individuals identifies known and novel cross-population and ancestry-specific associations as novel risk loci for Alzheimer's disease. *Genome Biol* 26, 210 (2025). <https://doi.org/10.1186/s13059-025-03564-z>
- Kuzma, Amanda, et al (2024). NIAGADS: A Comprehensive National Data Repository for Alzheimer's Disease and Related Dementia Genetics and Genomics Research, 8 Oct. 2024, <https://doi.org/10.1101/2024.10.07.24315029>.
- Naj, Adam., et al. (2023). Multi-Ancestry Genome-wide Association Analysis of Late-Onset Alzheimer's Disease (LOAD) in 60,941 Individuals Identifies Multiple Novel Cross-Ancestry Associations and Implicates Amyloid and Complement Pathways. *Alzheimer's & Dementia*. 19. 10.1002/alz.079401.

AWARDS & HONORS

Second place, 2025 Penn HealthCare Case Competition – *Blue Crane Bio Consulting*

2025

- Finalist team selected from **49 global applicants across 44 universities (including UPenn, MIT, Harvard, Yale, London Business School, and National University of Singapore)**; Developed commercialization and market-entry strategy for Müst Therapeutics, evaluating competitive positioning, reimbursement landscape, and growth opportunities in a healthcare-focused consulting competition